

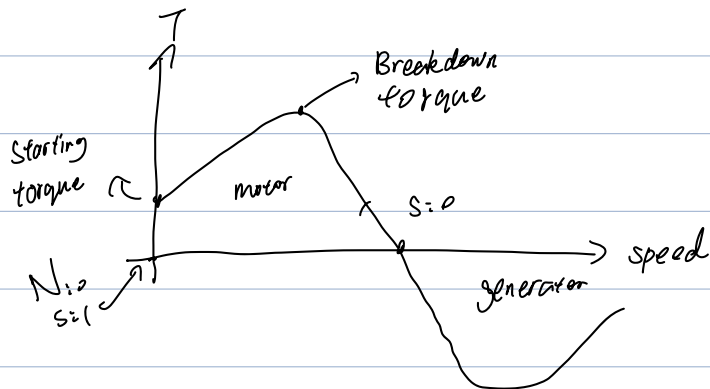
2023

Q2.

(a) Induction motor

- Industrial Fan
- Railway Traction
- Hoise
- Life

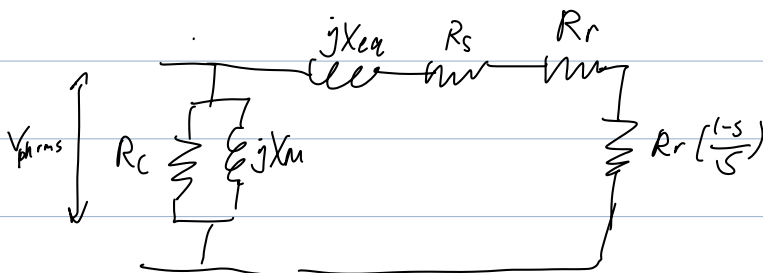
Induction generator : DFIG
on shore wind generator



$$(c) \quad n_s = \frac{120 f_s}{p} \Rightarrow f_s = \frac{p n_s}{120} = \frac{4 \times 1500}{120} = 50 \text{ Hz}$$

$$V_{rms} = 0.612 V_{dc} \cdot M_a \Rightarrow M_a = \frac{V_{Lrms}}{0.612 V_{dc}} = \frac{200}{0.612 \times 400} = 0.817$$

$$(d) \quad s = \frac{n_s - n_r}{n_s} = \frac{1500 - 1350}{1500} = 0.1$$



$$Z_T = (R_s + \frac{R_r}{s}) + jX_{eq} = 2.2 + 0.6j = 2.28 \angle 15.255^\circ$$

$$I_r = \frac{V_{Lrms}/\sqrt{3}}{Z_T} = 50.645 \angle -15.255^\circ$$

$$P_{mech} = 3 I_r^2 R_r \frac{1-s}{s} = 13.850 \text{ kW}$$

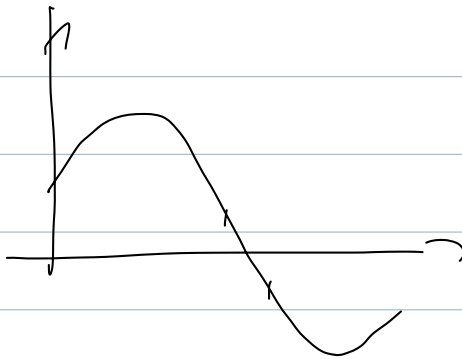
$$P_{loss} = 3 I_r^2 (R_r + R_s) + 3 \frac{V_{ph}^2}{R_c} = 3.745 \text{ kW}$$

$$T = \frac{P_{mech}}{\omega} = 97.97 \text{ N.m}$$

$$\text{eff}_i = \frac{13.850}{13.850 + 3.745} \times 100\% = 78.72\%$$

(E)

(i)



Assumption:

① The V curve is symmetrical around ω_s

② Machine eff remains at 78%.

③ Inverter eff 97%

$$0.05 = \frac{1500 - N_r}{1500} \Rightarrow N_r = 1425 \text{ rpm}$$

$$0.05 = \frac{N_r' - 1500}{1500} \Rightarrow N_r' = 1575 \text{ rpm}$$

$$T' = \frac{T}{2} = \frac{97.97}{2} = 48.99 \text{ N.m}$$

$$P_{mech} = T' \cdot N_r' = 8080.10 \text{ W}$$

$$P_{dc} = P_{mech} \times \text{Eff}_{gen} \times \text{Eff}_{inverter} = 6.17 \text{ kW}$$

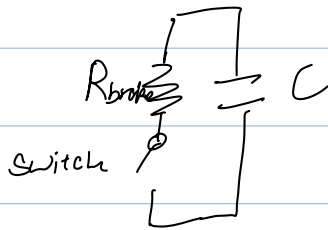
$$\int E_{440} = \frac{1}{2} C V^2 = \frac{1}{2} \cdot 20m \cdot 440^2 = 1936 \text{ J}$$

$$\int E_{400} = \frac{1}{2} C V^2 = \frac{1}{2} \cdot 20m \cdot 400^2 = 1600 \text{ J}$$

$$\Delta E = 1436 - 1100 = 336 \text{ J}$$

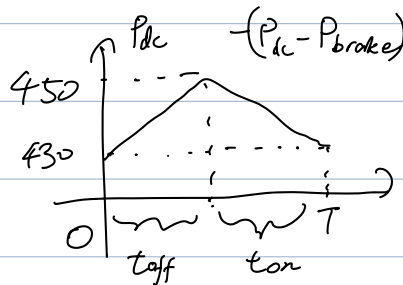
$$\Delta t = \frac{\Delta E}{P_{dc}} = \frac{336}{6.17k} = 54.46 \text{ ms}$$

(ii) The solution is to dissipate the extra energy into the resistor. However the PWM controller switch should be installed in series with the resistor to dump the energy in there when needed.



$$(i) P_{brake} = \frac{V_{avg}^2}{R_{brake}}$$

$$T = \frac{1}{f} = 0.1 \text{ s}$$



$$E_{450} = \frac{1}{2} C V_{450}^2 = 2025 \text{ J}$$

$$E_{430} = \frac{1}{2} C V_{430}^2 = 1844 \text{ J}$$

$$P_{dc} \cdot t_{off} = E_{450} - E_{430} = 176 \text{ J}$$

$$\Rightarrow t_{off} = \frac{176}{6169.2} = 28.53 \text{ ms}$$

$$t_{on} = 71.47 \text{ ms}$$

$$-(P_{dc} - P_{brake}) \cdot t_{on} = \Delta E$$

$$\Rightarrow P_{brake} = P_{dc} + \frac{\Delta E}{t_{on}} = 8.633 \text{ kW}$$

$$= \frac{V_{avg}^2}{R_{brake}}$$

$$\Rightarrow R_{brake} = 22.43 \Omega$$

$$f) \quad 30 \text{ kW} = \frac{1}{2} \rho \pi C_F \underline{V^3} R^2$$

$$\Rightarrow C_p = 0.2436$$

$$\Rightarrow T_{SR} = 5 = \frac{\omega R}{v}$$

$$\Rightarrow \omega = \frac{10}{8} \times 5 = 6.25 \text{ rad/s}$$

~~$$\Rightarrow f = \frac{1}{2\pi} = 29.84 \text{ Hz} \approx 30 \text{ Hz}$$~~

$$N_r \approx N_s = N_w \cdot \frac{60}{2\pi} = 1796.8 \text{ rpm}$$

$$N_s = \frac{120f}{p} \Rightarrow f = \frac{p N_s}{120} = \frac{4 \times 1796 \cdot 89}{120} = 6518$$

Q. 1. offshore $\rightarrow$ PM synchronous gen
(no gear box)
 $\downarrow$
higher power
 $\downarrow$
less than 15 MW

onshore $\rightarrow$ doubly fed
induction gen
 $\downarrow$
lower power
 $\downarrow$
less than 3 MW